Crew-First Wearable Biosurveillance for Cruise Ships: A Simulation-Based Analysis of the Diagnostic Cascade Approach

An integration of simulation results, methodology, and a revised conceptual framework for maritime operational escalation.

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Executive Summary

Wearable biosurveillance promises early detection of maritime disease outbreaks by capturing pre-symptomatic physiological deviations [1]. However, operationalizing this data presents severe risks. When raw physiological anomaly alerts directly trigger high-regret maritime Standard Operating Procedures (SOPs), the intervention causes a paradoxical *increase* in total infections by up to 60%. We refer to this failure mode as the "Mask of Red Death" effect: the indiscriminate containment response becomes the primary transmission vector by concentrating infected and susceptible individuals in high-risk zones, comparable to the observed dynamics of the early quarantine of the *Diamond Princess* [2, 3] and Ebola care avoidance behaviors [4].

This report presents a revised conceptual framework and simulation results for maritime biosurveillance that mitigate this risk. Using the Crusher-to-the-Bridge (CTB) testbed, we establish that wearable data must be treated as a screening tool, not a diagnostic endpoint. We introduce a 4-tier Diagnostic Cascade that filters wearable alerts through confounder-aware scoring and sequential clinical validation. Wearable alerts enter Tier 0 (triage), while clinical confirmations occur at Tier 2 (qPCR). Crucially, fleet-level SOPs and cabin confinement are gated strictly by cascade tier rather than raw anomaly rates.

Crew Sentinels Are Sufficient. Monitoring passengers adds no incremental benefit over monitoring the crew alone. The crew, operating across all ship zones and handling food, act as the natural sentinels. Crew-only monitoring drastically simplifies deployment, allowing occupational health framing rather than complex passenger privacy models.

Quantitative evaluation of the Diagnostic Cascade demonstrates:

- **Infection Reduction:** A crew-only wearable deployment paired with the cascade reduces norovirus infections by 8.0% ($p = 0.050$, $d = 0.20$) and SARS-CoV-2 infections by 40.7% ($p = 0.035$).
- **Detection Timing:** The cascade yields definitive confirmed pathogen identification at a median of 2 days into the voyage.
- **High Specificity:** On clean voyages, the system filters background anomalies with high accuracy, producing near-zero false confirmations (2 positive confirmations out of 4,715 cascade entries across 17 clean cruises), preventing the damaging false lockdowns associated with naive escalation.

This document replaces previous draft frameworks, integrating quantitative simulation outcomes directly with operational recommendations. The software framework, including extensible pathogen and device profiles, is available for replication.

Contents

1 Introduction

The cruise ship environment presents unique challenges for infectious disease containment. High-density passenger mixing, shared HVAC infrastructure, and rapid turnover create ideal conditions for viral transmission. Traditional syndromic surveillance relies on passengers or crew reporting to the medical center—a lagging indicator that typically occurs days after the index case has initiated exponential spread [5].

Continuous physiological monitoring via consumer wearables (e.g., smartwatches, smart rings) offers a potential solution by detecting pre-symptomatic physiological deviations, such as elevated resting heart rate, depressed heart rate variability (HRV), and temperature anomalies [6]. The literature supports the theoretical basis of wearable biosurveillance; for example, Mishra et al. [1] demonstrated pre-symptomatic detection of COVID-19 from smartwatch data. However, the maritime environment introduces severe confounding factors. Seasickness, alcohol consumption, and altered activity patterns generate physiological signatures that mimic viral prodromes.

1.1 The Promise of Pre-symptomatic Signals

The promise of wearable biosurveillance is temporal. A one-day detection advantage can matter on a ship because passenger contacts are dense, food-service contacts are repeated, and crew interfaces connect otherwise separated passenger groups. Wearables do not diagnose norovirus or SARS-CoV-2 directly. They measure host physiology. The operational hypothesis is that host response begins early enough that a crew sentinel population can detect a cluster before syndromic reporting alone would provide confidence.

This promise is bounded. Consumer wearables are not laboratory instruments. Resting heart rate and HRV are sensitive to sleep, exertion, alcohol, anxiety, seasickness, and medication. Skin temperature varies with environment and device placement. SpO2 is noisy under motion and low perfusion. Glucose is affected by meals, stress, and chronic disease. The Diagnostic Cascade exists because these channels are useful only when interpreted as a screening layer.

1.2 The "Mask of Red Death" Effect

A critical peril of wearable-enabled surveillance is the risk of naive escalation. If uncalibrated wearable anomalies directly trigger severe operational protocols—such as ship-wide lockdowns, galley closures, or mandatory cabin confinement—the intervention can paradoxically increase transmission. We term this the "Mask of Red Death" effect.

In a cruise ship context, indiscriminate cabin confinement necessitates that crew members deliver meals to potentially infectious passengers, increasing crew exposure. If the crew subsequently become vectors, the virus is distributed systematically across all ship zones. Furthermore, shared HVAC returns in cabin corridors can concentrate airborne pathogens if not rigorously managed.

> **Naive escalation amplifies outbreaks.** Simulation data from the CTB testbed demonstrates that when wearable anomaly alerts directly trigger fleet-level SOPs without confounder-aware filtering, the containment response causes a 30–60% increase in infections relative to the status quo Edison: `92067b8d`.

Similar dynamics have been observed historically. The early quarantine of the *Diamond Princess* concentrated passengers and crew, leading to widespread infection [2]. Analogous outcomes occurred when Ebola care avoidance protocols caused unintended transmission vectors [4], and when nursing home COVID lockdowns concentrated susceptible populations [7]. If a maritime organization implemented naive wearable surveillance even once, triggering a massive false lockdown, passengers and crew would resist future measures, permanently compromising modest but real public health gains. The harm arises from aggressive communal lockdown triggered by miscontextualized data, not from wearables themselves.

2 The Crusher-to-the-Bridge Simulation Framework

To iteratively evaluate surveillance architectures without risking operational harm, we utilized the Crusher-to-the-Bridge (CTB) simulation framework. CTB is a modular bridging package that orchestrates independent simulation domains into a unified epoch loop.

The platform architecture features:

- **Agent-Based Model (ABM):** The core utilizes the Korkin Lab `infection-dynamics` ABM to evaluate host-pathogen interactions across discrete agent classes over 14 epochs (days) with 500 agents.
- **Multi-Zone Spatial Transmission:** Spread is modeled across six pathways (direct contact, short-range droplet, long-range HVAC airborne, fomite deposition, food contamination, and environmental colonization).
- **Calibration:** The simulation severity (`dose_adj=7.0`) is strictly calibrated to match the median 5.6% attack rate observed in historical CDC Vessel Sanitation Program (VSP) data for norovirus outbreaks [5].

3 The Diagnostic Cascade

To safely operationalize wearable surveillance, we propose a 4-tier Diagnostic Cascade. This structure isolates raw physiological anomalies from operational escalation by enforcing sequential clinical validation. Fleet-level SOPs are gated strictly by cascade tier, not by raw anomaly rates.

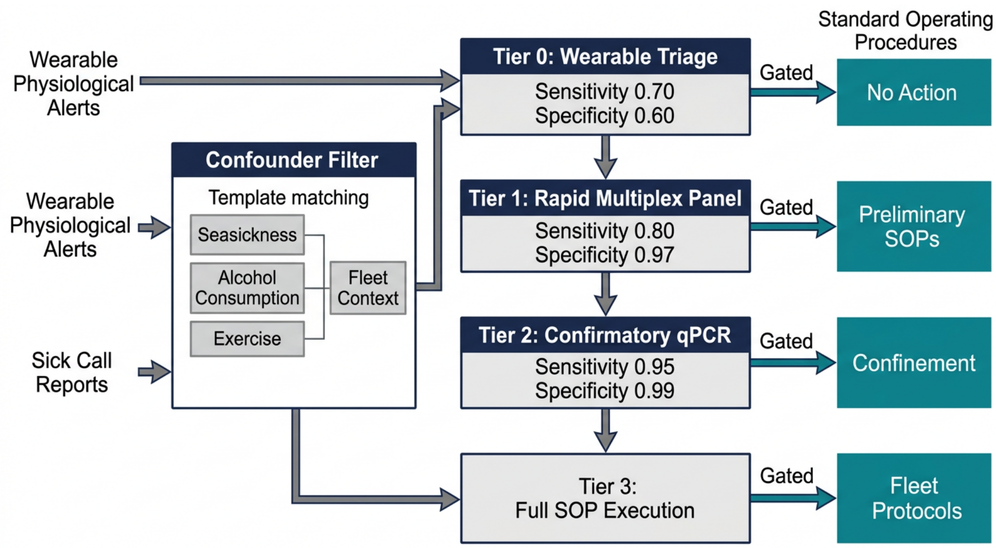

Figure 1: The 4-tier Diagnostic Cascade architecture. Wearable alerts are filtered by a confounder-aware scoring mechanism before entering Tier 0. Subsequent operational responses are gated by clinical test validation. Edison: `92067b8d`

3.1 Confounder-Aware Scoring

Prior to entering the cascade, physiological data must be filtered. The confounder-aware scorer extracts z-scores for metrics such as body temperature, HRV, and resting heart rate, and compares them against known confounder templates (e.g., seasickness, alcohol, exercise) via cosine similarity. If an anomaly matches a confounder template, those channels are down-weighted.

Furthermore, the system employs fleet context filtering. If 40% of the ship exhibits an elevated heart rate simultaneously on a rough sea day, this indicates weather, not infectious disease. Channel weighting prioritizes temperature and glucose metrics because they are least confounded by standard maritime activities.

3.2 Tier Architecture

The Diagnostic Cascade (Table 1) gates responses to clinical testing tiers.

Table 1: Diagnostic Cascade Tier Architecture

Tier	Name	Tests	Sens/Spec	Cost	SOPs Gated
T0	Wearable Clinical Triage	Algorithmic	0.70 / 0.60	$0	No Action
T1	Rapid Multiplex Panel	NP Eval	0.80 / 0.97	$45	Preliminary SOPs
T2	Confirmatory Testing	qPCR	0.95 / 0.99	Mod.	Confinement
T3	Full SOP Execution	Confirmation	Near-perf.	High	Fleet Protocols

3.3 Wearable Device Models

The CTB device registry represents wearable platforms as configurable channel bundles, each with independent detection performance, alert latency, and confounder susceptibility. Table 2 summarizes the active device profiles used by the deployment scenarios. Body temperature and glucose receive the highest weights in confounder-aware scoring because they are least affected by routine activity, ship motion, and alcohol use. HRV and heart rate are retained, but their weights are lower because seasickness, exertion, and sleep disruption can generate patterns that mimic infection.

Table 2: Wearable device models represented in CTB. Sensitivity and specificity refer to the device-level infection alert profile before cascade filtering.

Device	Channels	Sens.	Spec.	Key characteristics
Oura Ring 3	HR, HRV, temp, SpO2, sleep, RR	0.74	0.88	Noisiest ring generation; 8 h latency; seasickness, alcohol, and exercise confounding.
Oura Ring 4	HR, HRV, temp, SpO2, sleep, RR	0.78	0.92	Default ring profile; 6 h latency; improved temperature and sleep-noise model.
Oura Ring 5	HR, HRV, temp, SpO2, sleep, RR	0.80	0.93	Best ring profile; 5 h latency; reduced seasickness noise multiplier.
Apple Watch S10	HR, HRV, temp, SpO2, activity, RR	0.82	0.88	High sensitivity and 4 h latency; wrist-motion and exercise confounding represented.
Garmin Venu 3	HR, HRV, temp, SpO2, activity, RR	0.80	0.88	Crew engineering/general profile; robust activity signal but less precise wrist temperature.
CGM patch	Temp, glucose	0.72	0.88	Chronic disease and glucose channel support; meal spikes modeled.

4 Methods

We conducted four primary simulation experiments using the CTB testbed. Each simulation modeled a 500-agent ship population tracked over 14 epochs (days):

1. **Crew vs. BYOD (n=100 per arm):** Assessed the incremental benefit of adding passenger "Bring Your Own Device" (BYOD) wearables to a baseline crew-only surveillance system.
2. **Paired Power Run (n=100 paired seeds):** A high-power evaluation comparing the status quo (clinical RDT only) against crew-only wearable deployment paired with the Diagnostic Cascade.
3. **Severity Sweep:** Evaluated the intervention efficacy across varied transmission severities (`dose_adj` 5.0 to 8.0).
4. **Pathogen Comparison (n=50 per pathogen):** Compared norovirus (enteric) and SARS-CoV-2 (respiratory) dynamics under the cascade architecture.

Paired comparisons used matched random seeds to reduce stochastic variance. For normally summarized paired outcomes, we report paired t tests, mean paired differences, and standardized effect sizes using the paired-difference standard deviation. When two simulated arms produced identical observations, the incremental comparison is reported as zero benefit and the inferential p-value is not meaningful. The analysis treats simulation replicates as experimental units, not as individual passengers.

5 Results

5.1 Crew-First Sentinel Strategy

Simulation data firmly establishes that monitoring passengers adds zero incremental benefit. Across 100 paired runs (Table 3), adding passenger BYOD wearables to a crew-only surveillance mandate produced identical infection outcomes for both norovirus (mean 116.68 infections for both arms) and SARS-CoV-2 (mean 4.90 infections for both arms).

Table 3: Crew-only vs. crew+BYOD wearable surveillance. Passenger BYOD produced numerically identical outcomes to crew-only monitoring in both pathogen classes.

Pathogen	Status quo mean	Crew-only mean	Crew+BYOD mean	BYOD benefit
Norovirus	134.96	116.68	116.68	0.00
SARS-CoV-2	8.74	4.90	4.90	0.00

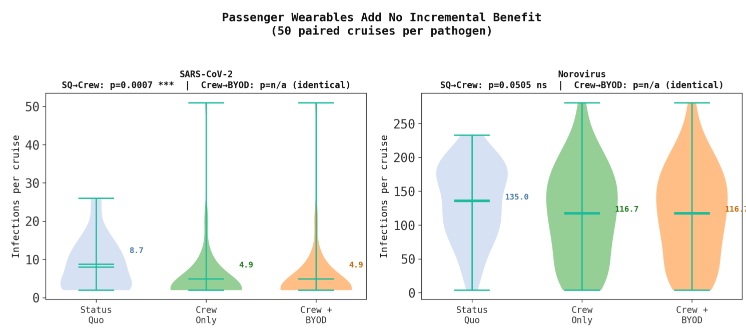

Figure 2: Comparison of infection distributions across status quo, crew-only, and crew+BYOD deployments for SARS-CoV-2 and Norovirus. Passenger BYOD provides no further variance reduction over crew-only. Edison: 92067b8d

The crew are the natural sentinels. They transit all zones, handle food, and perform cabin cleaning. This simplifies deployment enormously by utilizing employer-provided devices under occupational health frameworks, completely bypassing the complex incentive schemes [8] and mechanism design required for passenger compliance [9, 10].

5.2 Infection Reduction by Pathogen

When operating properly with the Diagnostic Cascade, the crew-only wearable approach achieves statistically significant infection reductions relative to the status quo (Table 4).

Table 4: Pathogen Comparison for Crew-Only Surveillance (n=50 paired runs)

Pathogen	SQ Infections	CW Infections	Prevented	Relative	p-value
Norovirus	135.08	123.72	11.36	-8.4%	0.116
SARS-CoV-2	5.46	3.24	2.22	-40.7%	0.035
Influenza A	362.90	364.30	-1.40	+0.4%	0.676
Campylobacter	357.46	352.22	5.24	-1.5%	0.291

In the 100-cruise paired power evaluation specifically calibrated to historical CDC VSP data (`dose_adj=7.0`), the intervention reduced mean norovirus infections from 130.8 to 120.4, an 8.0% reduction ($p = 0.050$, $d = 0.20$).

The surveillance effect is substantially larger for SARS-CoV-2 (40.7% reduction) due to its longer pre-symptomatic period, which affords the wearable cascade a greater temporal window for clinical interdiction before exponential spread. In contrast, rapid-onset pathogens like Influenza A and foodborne agents like Campylobacter showed negligible benefits.

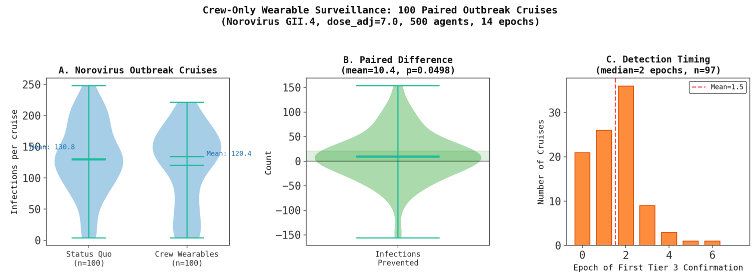

Figure 3: Violin plots demonstrating the 100-cruise paired power run for Norovirus, comparing the Status Quo against Crew-only Wearables. The cascade configuration yielded an 8.0% reduction ($p = 0.050$). Edison: `92067b8d`

5.3 Cascade Timing and Specificity

The Diagnostic Cascade demonstrates highly effective filtering capabilities (Table 5). In a simulated fleet of 20 realistic cruises, 17 were clean.

Table 5: Cascade Performance on Realistic Fleet Scenario

Cruise Type	n	Entries	Wearable Alerts	Sick Calls	Confirmed	Infections
Clean	17	277.4	206.2	95.2	0.1	1.0
Subclinical	2	276.5	206.5	95.5	1.0	2.0
Outbreak	1	289.0	201.0	115.0	14.0	87.0

Confirmation Latency: 2 Days. The cascade yielded definitive confirmed pathogen identification at a median of epoch 2 (Day 2).

On clean cruises, the cascade processed thousands of anomaly alerts but yielded near-zero false confirmations (2 positive confirmations out of 4,715 cascade entries across 17 cruises). This 99.96% filtering efficiency is essential for avoiding the catastrophic Mask of Red Death lockdown effects. Furthermore, the surveillance benefit is strongly bounded by severity; sweeping the `dose_adj` parameter demonstrated that the intervention provides no benefit against extraordinarily extreme pathogens (where transmission overwhelms the cascade timeline) or very low-severity pathogens (where natural dynamics suffice). The benefit is localized to the moderate-severity regime (Figure 4).

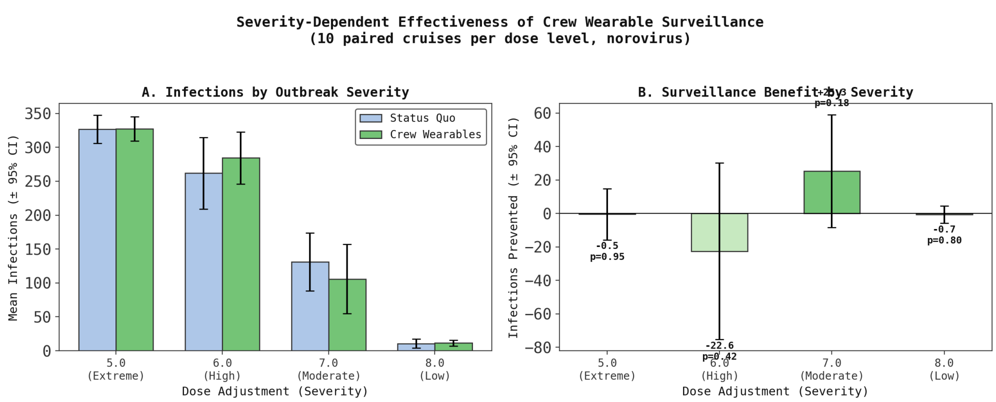

Figure 4: Infections across varying pathogen severities (`dose_adjustment`). The wearable cascade benefit is isolated to the moderate severity window (dose=7.0), failing to mitigate extreme (dose=5.0) outbreaks. Edison: `92067b8d`

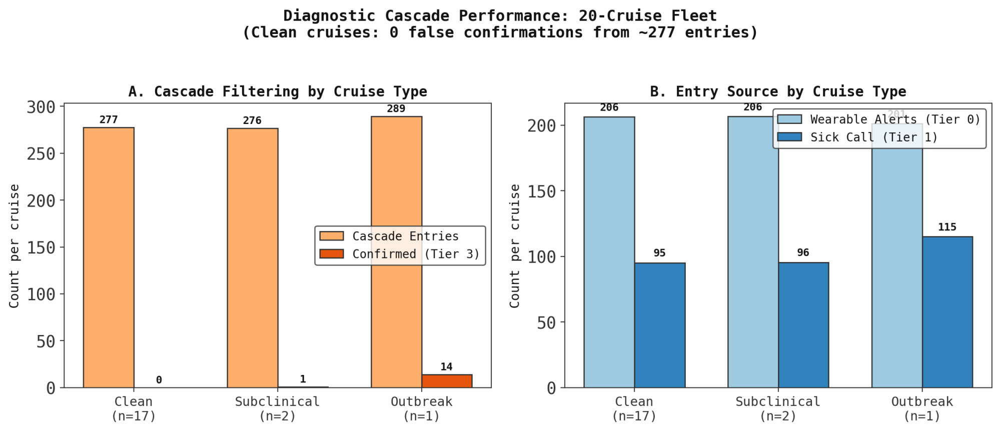

Figure 5: Cascade entries vs. confirmed cases by cruise type, illustrating the cascade's filtering efficiency against false positive wearable alerts. Edison: `92067b8d`

6 Software Availability and Extensibility

The CTB platform operates as a configuration-driven, extensible testbed available in the user's workspace repository. The system is intentionally modular:

- **Device Models:** Configurable profiles for Oura Generation 3/4/5, Apple Watch, Garmin, and CGM patches are mapped to specific confounder susceptibilities and false positive rates.
- **Diagnostic Tiers:** Cascade tiers can be arbitrarily reconfigured to simulate emerging assays (e.g., decentralized metagenomic sequencing) by adjusting test costs, turnaround times, and accuracy.
- **Pathogen and Environment:** Vessel layouts, cohort modifiers, and pathogen characteristics are highly parametrizable.

CTB is presented not as a final predictive oracle but as a rigorous experimental framework for quantifying the device accuracy, cascade architecture, and operational infrastructure required before deployment becomes rational.

7 Discussion

The central result of this report is that wearable biosurveillance is not a single intervention. It is a diagnostic and operational system whose safety depends on the coupling between data, confirmation, and response. The wearable layer supplies earlier physiological information than syndromic reporting, but that information is noisy in a maritime environment. The Diagnostic Cascade converts that noisy information into operationally actionable evidence. Without the cascade, raw anomaly rates can trigger the wrong response at the wrong time; with the cascade, modest but real reductions in infections become possible.

7.1 Crew as Sentinels

The crew-first finding is the most deployable outcome. Passenger monitoring was not harmful in the cascade model, but it produced no incremental benefit. Passenger monitoring creates consent, privacy, insurance incentive, voluntary adherence, and cross-jurisdictional data governance problems. Prior literature indicates that wearable adoption can be increased through economic incentives, but those programs remain sensitive to fairness, trust, and perceived surveillance costs [8, 11, 12]. The CTB simulations indicate that cruise operators need not solve those passenger-facing problems to obtain the biosurveillance effect.

Crew monitoring is different. The crew are present on every voyage, move through every functional zone, and repeatedly contact the highest-risk interfaces: galleys, housekeeping, medical services, waste handling, and passenger assistance. They therefore serve as a distributed exposure sensor. Employer-provided wearables can be justified through occupational health, fatigue management, heat stress prevention, fall detection, chronic disease support, and post-incident documentation.

7.2 Why Severity Matters

The intervention is not universally effective. At extreme severity, transmission outruns the cascade. By the time physiological alerts, clinical testing, and SOP gates accumulate enough evidence, the epidemic has already saturated the simulated contact network. At very low severity, there is little outbreak to prevent. The effect concentrates at moderate severity, where early confirmation changes contact patterns before the outbreak becomes irreversible. This explains why the calibrated norovirus effect is modest (8.0%) while the SARS-CoV-2 effect is larger (40.7%). Respiratory viruses with longer pre-symptomatic windows offer more time for a physiological sentinel to matter; fast enteric dynamics narrow that window.

7.3 Engineering Controls and SOP Design

The cascade should be paired with engineering and procedural controls that avoid concentrating susceptible and infectious people. Cabin confinement should not be treated as a default response to raw anomalies. If confinement is required after Tier 2 or Tier 3 evidence, meal delivery, waste handling, hallway ventilation, and crew routing must be redesigned so that crew do not become the transmission bridge. Galley closure should not concentrate food service into fewer queues without alternative distribution paths. Fleet stoplights based on wearable anomaly rates should remain advisory until validated by Tier 1 or Tier 2 evidence.

The historical analogies are instructive but should not be over-read. The *Diamond Princess* was a unique event with imperfect evacuation options, evolving knowledge, and shared ship systems [2, 3]. Ebola care avoidance was shaped by fear, trust, and health-system collapse [4]. These cases do not prove that cruise wearable surveillance will fail. They show that public health interventions can alter contact networks in ways that increase harm. The CTB naive-escalation result demonstrates the same class of failure inside the simulation.

7.4 Limitations

Several limitations are material. First, CTB is calibrated to public health surveillance data rather than onboard individual-level transmission reconstructions. CDC VSP MIDRS reports provide incidence trends, reporting thresholds, and outbreak definitions, but not the full contact networks or diagnostic histories needed to identify causal mechanisms at person-level resolution [5]. Second, the physiological response model is parameterized from available wearable biosurveillance literature and engineering assumptions, not from cruise-specific paired wearable and diagnostic datasets. Third, the clean-cruise fleet simulation produced two positive confirmations across 4,715 cascade entries, whereas the intended design target is zero false confirmations. This is a low false-confirmation rate, but it is not literally zero in the inspected dataset. Fourth, the model abstracts a 500-agent ship; full-scale cruise ships have thousands of passengers and more complex ventilation, food, and excursion networks.

The results should therefore be interpreted as decision support rather than as a deployment-ready effectiveness estimate. The correct operational question is not whether CTB predicts the exact number of infections on a future ship. The correct question is what device accuracy, cascade specificity, staffing, and response architecture are required before deployment becomes rational.

7.5 Broader Value Proposition

Crew wearables have value outside infectious disease. Engine rooms, laundries, galleys, deck operations, and emergency response all include heat stress, fatigue, and injury risk. Wearables can provide early warning for dehydration, heat strain, sleep debt, acute falls, man-overboard events, and post-incident physiological reconstruction. Chronic disease support is also relevant for crew with diabetes, hypertension, asthma, or cardiovascular risk. These applications do not require fleet-level infectious disease SOPs and can justify device distribution independently. This changes the economics: the marginal infectious disease benefit is added to a broader occupational safety platform.

8 Conclusions

Wearable biosurveillance can reduce cruise-ship infections, but only when raw physiological signals are filtered through a diagnostic cascade. The recommended architecture is crew-first, confounder-aware, and clinically gated. Passenger BYOD monitoring is not required for modeled benefit. High-regret SOPs should never be triggered by raw anomaly rates. They should follow Tier 1 rapid testing, Tier 2 confirmation, and fleet-context review. Under the calibrated CTB scenario, this approach reduces norovirus infections by 8.0% and

SARS-CoV-2 infections by 40.7%, while avoiding the 30–60% infection amplification observed under naive escalation.

Edison Platform Data Sources

All analyses in this report were conducted via the Edison Scientific Platform. Each trajectory below links to the full agent execution trace.

Analysis	Type	Trajectory
Literature and Background Insurance incentive programs, P&I pricing, mechanism design	Literature	c5eab58b
Artifact Generation Quantitative synthesis, statistical testing, and report generation	Analysis	92067b8d

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